A NEEDLE IN FIVE TINY HAYSTACKS: LOCAL LEARNING COEFFICIENTS OF SCALAR RESNETS

BOBBY FABER

Abstract. A residual network can represent one function at many points of its parameter space, and singular learning theory assigns each point its own local learning coefficient λ and multiplicity m. We compute both exactly for the simplest nontrivial case: the scalar network obtained by composing L maps $x \mapsto x + w^r x^r$, with squared-error loss in the raw weights w. A depth-five cubic network represents $f(x) = x + x^3$ with its single nonzero weight in any of five positions, yet the local pair is $(13/12, 1)$, $(7/6, 2)$ or $(7/6, 1)$ depending on the position. In general, the nonzero (active) blocks split the remaining weights into consecutive inactive groups. For odd r the weights within a group can cancel. In every case λ is a regular active contribution plus one identity-target contribution per group, and for odd r a logarithm appears when a group has length exactly r. We also classify every target fiber, count its points exactly, and determine the global pair. Lean certifies the pairs, fibers and counts of all three theorems for localizations on small balls with bounded measurable densities, under the stated Gram assumption.

1. Introduction

A depth-five cubic residual network represents $f(x) = x + x^3$ with its sole nonzero weight at any of its five positions, yet these realizations have different local loss geometry. We describe this geometry by a pair (λ, m): the local learning coefficient λ and multiplicity m give the order $N^{-\lambda}(\log N)^{m-1}$ of the local integral of e^{-NK_f}, where K_f is the squared-error population loss and $N \to \infty$ is a positive concentration parameter. At positions 1 or 5, the pair is $(13/12, 1)$; at positions 2 or 4, it is $(7/6, 2)$; at the center, it is $(7/6, 1)$. Thus the same function has local integral orders $N^{-13/12}$, $N^{-7/6} \log N$, and $N^{-7/6}$. Figure 1 and Table 1 display the three cases. In the terms of our title, the needle is the same in all five realizations: one active block computing $x \mapsto x + x^3$. What differs is the haystack, the inactive blocks packed around it, and the arrangement theorem below shows that the haystack's shape alone decides how the learning coefficient differs.

The local learning coefficient is by definition an invariant of a point in parameter space, not of the function computed there (Watanabe, 2009; Lau et al., 2025), and it captures singular geometry beyond parameter counting or Jacobian rank (Wei et al., 2023). The example makes both facts exact. Every realization has five parameters and raw coefficient-Jacobian rank one, yet the local pairs differ, and the multiplicity varies within a single fiber. A local analysis near one realization therefore need not describe another realization of the same function, and which realization a training process approaches changes what a local estimate measures.

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The mechanism is cancellation. A block with zero weight is the identity map, so moving the single active block along the chain does not change the output. Perturbing the inactive weights does change it, and for odd r the leading terms are signed rth powers that can cancel against each other. Cancellation happens only among consecutive inactive blocks. An active block between two groups composes their contributions with a nonlinear map, which places them in independent leading loss directions (Appendix C). The positions of the active blocks therefore decide which weights can cancel together. A group's contribution changes regime as its length passes r; at length exactly r the two regimes balance and a logarithm appears. Theorem 2 adds these group contributions to the regular contribution of the active weights. Theorem 3 classifies every target fiber and selects its global pair: local pairs depend on the inactive-group lengths, while the global pair depends only on degree, depth, and active count.

The architecture is the scalar monomial specialization of the polynomial ResNets studied by Dahlgren et al. (2026). Related work connects sparse subnetworks to singularities of polynomial function spaces (Shahverdi et al., 2026), and Kohn et al. (2026) relate rank-deficient critical points of deep monomial networks to subnetworks with inactive neurons. Our realizations are subnetworks of exactly this kind, and they show that equal Jacobian rank does not determine the learning coefficient. Here we compute local loss-volume exponents, giving exact examples relevant to local learning coefficient estimation (Lau et al., 2025). Exact coefficients are also known for reduced-rank regression and multilayer linear networks (Aoyagi and Watanabe, 2005; Aoyagi, 2024). Our calculations use the singular learning framework and monomial rules of Watanabe (2009); Lin (2017).

The model is deliberately minimal: one dimension, no biases and monomial activations. We chose it because it is exactly solvable, so every learning coefficient can be computed and certified rather than estimated; it is not a model of practical networks. Within it, the theorems give exact local and global pairs for every degree, depth and active count. They show that the learning coefficient depends on the realization, not only on the function, explain the dependence by cancellation within inactive groups, and place the odd-degree transition exactly where Newton nondegeneracy fails (Section 3). All statements concern deterministic population-loss integrals with priors in raw weights. Finite samples, stochastic estimators such as that of Lau et al. (2025), and networks with width or biases are outside their scope.

2. Model and assumptions

Fix integers $r \geqslant 2$ and $L \geqslant 1$. The parameters are independent raw weights $w \in \mathbb{R}^L$, with

$$h_0(x) = x, \qquad h_\ell(x) = h_{\ell-1}(x) + w_\ell^r h_{\ell-1}(x)^r, \quad 1 \leqslant \ell \leqslant L. \tag{1}$$

This is the bias-free scalar specialization of $h \mapsto h + \sigma(wh)$ with $\sigma(z) = z^r$ (Dahlgren et al., 2026, Definition 2.1), hence the coefficient w^r. The weights are raw: priors are densities in w, not in $c = w^r$. This choice changes the answer, because $w \mapsto w^r$ is not a local diffeomorphism at zero, and a smooth prior in c corresponds to a raw density proportional to $|w|^{r-1}$, which vanishes

there. Write $F_c(x) = x + cx^r$, so $h_L = F_{w_L^r} \circ \cdots \circ F_{w_1^r}$. The target f is realizable in this architecture. For an input probability measure μ, let

$$K_f(w) = \int_{\mathbb{R}} (h_L(w, x) - f(x))^2 \, d\mu(x). \tag{2}$$

Put $\nu = (r^L - 1)/(r - 1)$ and $d_j = 1 + j(r - 1)$ for $1 \leqslant j \leqslant \nu$. Throughout, assume the sparse feature Gram matrix

$$G_{ij} = \int_{\mathbb{R}} x^{d_i + d_j} \, d\mu(x) \quad (1 \leqslant i, j \leqslant \nu) \quad \text{is finite and positive definite.} \tag{3}$$

Then K_f is comparable to the squared coefficient-error norm. A finite absolute moment of order $2r^L$ and infinite support of μ suffice: a nonzero univariate polynomial has finitely many zeros.

At a realization w^*, let π be a smooth positive density on a full neighborhood. Take a nonnegative smooth compact cutoff χ, supported in a sufficiently small neighborhood and equal to one near w^*. Define the local pair by

$$Z_{N,w^*} = \int_{\mathbb{R}^L} \chi(w) e^{-N K_f(w)} \pi(w) \, dw \asymp N^{-\lambda} (\log N)^{m-1}, \qquad N \longrightarrow \infty. \tag{4}$$

The symbol $\asymp$ denotes bounds by positive constant multiples. Equivalently, the local volume of $\{K_f \leqslant \epsilon\}$ has order $\epsilon^\lambda (\log(1/\epsilon))^{m-1}$ as $\epsilon \downarrow 0$. A measurable density bounded above and away from zero already suffices for this order; local constants may depend on r, L, w^* and μ. For every loss treated below, λ is also the local integrability threshold of K_f^{-s}, without resolution (Appendix B). For this analytic loss, m is the order of the first positive pole of the local zeta integral, by the usual resolution framework (Lin, 2017, Theorem 2.10 and Proposition 3.2); we use this only as an interpretation.[1] Global statements use a nonnegative measurable density on $\mathbb{R}^L$ with finite mass, continuous and strictly positive at every point of the target fiber.

3. The identity-target phase transition

Write $K_0 = K_x$ and $u_i = w_i^r$, and set

$$S(w) = \sum_{i=1}^{L} w_i^r, \qquad Q(w) = \sum_{i=1}^{L} w_i^{2r}.$$

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Theorem 1 (Identity target). *Under* (3)*, near* $w = 0$*,*

$$K_0(w) \asymp S(w)^2 + Q(w)^2. \tag{5}$$

The local pair is

$$(\lambda, m) = \begin{cases} (L/(2r), 1), & r \text{ even or } L < r, \\ (1/2, 2), & r \text{ odd}, \ L = r, \\ ((L + r)/(4r), 1), & r \text{ odd}, \ L > r. \end{cases} \tag{6}$$

Moreover, K_0 *is coercive and has zero set* $\{0\}$*. Thus the same pair governs the global integral under a finite-mass prior continuous and positive at zero.*

[1] Because our loss is squared, Lin's coefficient-ideal threshold is 2λ; his zeta convention is $(\sum_i f_i^2)^{-z/2}$.

WHAT IS CLASSICAL. Two comparisons separate the new part of Theorem 1 from the standard one; for even r the pair is immediate (see below). First, for odd r the leading term S^2 alone already has pair $(\min\{1/2, L/(2r)\}, m)$ with $m = 2$ exactly when $L = r$: S is homogeneous with zero set smooth away from the origin, and the blow-up of Appendix B without the Q^2 term gives a transverse exponent $1/2$ and a radial exponent $L/(2r)$. Theorem 1 therefore agrees with this classical threshold for $L \leqslant r$, logarithm included. For $L > r$ the radial term Q^2 cuts off the cone $\{S = 0\}$ and raises λ from $1/2$ to $(L+r)/(4r)$. Second, with a smooth positive prior in effective coordinates $u = w^r$, the loss is comparable to $(\sum_i u_i)^2 + \|u\|^4$, whose pair is $(1/2 + (L-1)/4, 1)$ for every r and L. The parity split, the critical length and the logarithm all come from the raw parametrization.

The first two output error coefficients are exactly S and

$$B = r \sum_{i<j} u_i u_j = \frac{r}{2}(S^2 - Q).$$

All higher coefficients have higher homogeneous degree in u. The Gram bound and this identity give (5); Appendix A supplies the uniform estimate and coercivity argument.

For even r, the summands of S are nonnegative, so $S^2 \asymp \|w\|^{2r}$ and the radial exponent is $L/(2r)$. For odd r and $L \geqslant 2$, write $w = \rho\theta$ with $\theta \in \mathbb{S}^{L-1}$. The angular function $\tau(\theta) = \sum_i \theta_i^r$ has a nonempty regular zero set. Near it, transverse coordinates $t = \tau(\theta)$ reduce the loss and volume to

$$\rho^{2r}(t^2 + \rho^{2r}), \qquad \rho^{L-1}\, d\rho\, dt\, d\eta \tag{7}$$

up to bounded positive factors. A finite collection of local charts with bounded positive densities suffices. In the thin region $|t| \leqslant \rho^r$, cancellation changes the radial scale to ρ^{4r} while the angular width contributes ρ^r. This gives $(L+r)/(4r)$. The competing exponent is $L/(2r)$; their crossing at $L = r$ produces one logarithm. Appendix B computes both exponents and the logarithm using two explicit sectors.

REMARK (NEWTON POLYHEDRA). The degree-$2r$ part of K_0 is $G_{11}S^2$ and all other terms have higher degree, so the Newton polyhedron of K_0 at zero is $\{\alpha \in \mathbb{R}^L_{\geqslant 0} : \sum_i \alpha_i \geqslant 2r\}$ and its Newton prediction (Varchenko, 1976; Arnold et al., 1988) is $(L/(2r), 1)$. The prediction is always an upper bound for λ, since $S^2 \leqslant LQ$ gives $K_0 \leqslant C \sum_i w_i^{2r}$. For even r, or $L = 1$, the face polynomial $G_{11}S^2$ has no zeros in $(\mathbb{R}^*)^L$, in particular no singular ones, and the prediction is exact. For odd r and $L \geqslant 2$ it vanishes to second order on $\{S = 0\} \cap (\mathbb{R}^*)^L$, which contains $(1, \dots, 1, -(L-1)^{1/r})$, so K_0 is Newton-degenerate. The degeneracy is invisible for $L < r$, where the cancellation exponent $(L+r)/(4r)$ exceeds $L/(2r)$; it costs the logarithm at $L = r$, and lowers λ to $(L+r)/(4r)$ for $L > r$.

4. The local arrangement theorem

Each inactive group inherits the identity-target geometry of Theorem 1; the active weights contribute regular directions. For an inactive group of size n, define $A_r(0) = 0$, $M_r(0) = 1$, and, for $n \geqslant 1$,

$$A_r(n) = \begin{cases} n/(2r), & r \text{ even}, \\ n/(2r) - (n-r)_+/(4r), & r \text{ odd}, \end{cases} \qquad M_r(n) = \begin{cases} 2, & r \text{ odd and } n = r, \\ 1, & \text{otherwise}. \end{cases} \tag{8}$$

Here $(a)_+ = \max(a, 0)$. These are the identity pairs of Theorem 1, with the empty group contributing an integral equal to one.

Fix w^* with exactly k nonzero weights, at positions $i_1 < \cdots < i_k$. Put $c_j = (w^*_{i_j})^r \neq 0$. There are $k+1$ consecutive inactive groups, of lengths $n_0, \ldots, n_k$, before, between, and after the active blocks; $k + \sum_g n_g = L$. Write their raw weights as $a_g = (a_{g,1}, \ldots, a_{g,n_g})$ and put

$$S_g = \sum_{i=1}^{n_g} a_{g,i}^r, \qquad Q_g = \sum_{i=1}^{n_g} a_{g,i}^{2r}.$$

The active deviations $\delta_j = w_{i_j}^r - c_j$ are valid local coordinates because $r(w^*_{i_j})^{r-1} \neq 0$.

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Theorem 2 (Local arrangement). *Let $f = h_L(w^*, \cdot)$ and assume (3), with any nonzero active values and any inactive group sizes $n_g \geqslant 0$. For odd $r \geqslant 3$, on a sufficiently small full neighborhood of w^*,*

$$K_f \asymp \sum_{j=1}^{k} \delta_j^2 + \sum_{g=0}^{k} (S_g^2 + Q_g^2). \tag{9}$$

For even $r \geqslant 2$ the comparison is

$$K_f \asymp \sum_{j=1}^{k} \delta_j^2 + \sum_{\text{inactive } i} |w_i|^{2r}. \tag{10}$$

Under the local prior conditions of Section 2, for every $k \geqslant 0$ with total depth $L \geqslant 1$,

$$\boxed{\lambda = \frac{k}{2} + \sum_{g=0}^{k} A_r(n_g), \qquad m = 1 + \sum_{g=0}^{k} \big(M_r(n_g) - 1\big).} \tag{11}$$

Independence of the retained directions follows by induction on active count: each new active block multiplies old directions by a polynomial coprime to the new active direction, while the new group directions have strictly higher degree.

Independence controls the retained coefficient directions, but cancellations can make those coefficients much smaller than the raw perturbation. A remainder estimate solely in the raw distance would then be insufficient: the omitted terms must be small relative to the retained coefficients themselves. Appendix C proves $\|E\| \leqslant C\varepsilon \|z\|$, where z collects the retained active and

group coefficients and $\varepsilon \to 0$ throughout the raw neighborhood. Absorbing this error proves the loss comparisons. Their model integral factors on product boxes: active directions contribute $k/2$, and group logarithmic powers add. Bounded positive prior and Jacobian factors preserve the result.

5. Same function, different local geometry

Take $r = 3$ and $f(x) = x + x^3$. Its complete fiber consists of the unit coordinate vectors e_j, $1 \leqslant j \leqslant L$: a composition with q nonzero blocks has degree 3^q, so there is exactly one active coefficient, which must equal one. Figure 1 shows how placement partitions the inactive weights. Table 1 applies (11) to all those points at depths four and five.

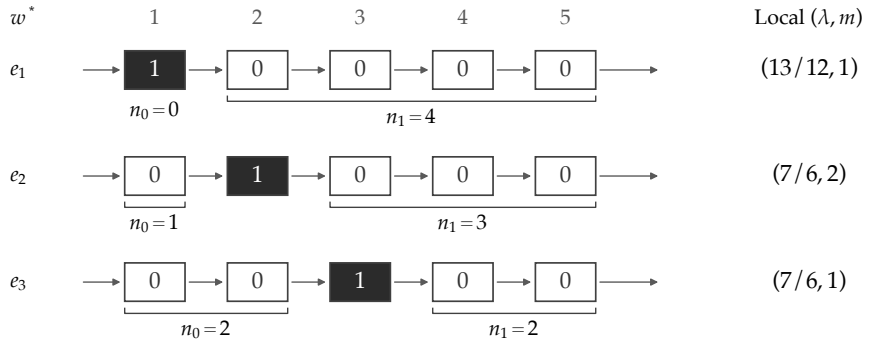


Figure 1. The three distinct local pairs for the depth-five cubic target $f(x) = x + x^3$. Boxes denote full residual blocks, labeled by raw weight; black boxes are active, and brackets mark consecutive inactive groups. The endpoint, one-position-inward, and center placements have group lengths $(0, 4)$, $(1, 3)$, and $(2, 2)$, respectively. Only the second has a group at the critical length $r = 3$, giving $m = 2$. By (11), positions 5 and 4 have the same pairs as 1 and 2; see Table 1.

Depth L	Active position j	Inactive lengths (n_0, n_1)	Local (λ, m)
4	1, 4	$(0, 3), (3, 0)$	$(1, 2)$
4	2, 3	$(1, 2), (2, 1)$	$(1, 1)$
5	1, 5	$(0, 4), (4, 0)$	$(13/12, 1)$
5	2, 4	$(1, 3), (3, 1)$	$(7/6, 2)$
5	3	$(2, 2)$	$(7/6, 1)$

Table 1. The same cubic target at every listed realization. Positions are counted in composition order.

At depth four, moving the active block changes the logarithmic factor. At depth five, it can change the exponent itself. All these points have raw coefficient-Jacobian rank one; that rank does not determine either pair. The

theorem also permits more than one logarithm: two groups of length r give $m = 3$. Such a local statement need not describe the globally dominant realization.

Remark (what a local learning coefficient sees). Let $\varpi = \chi\pi\,dw$ be the local measure of (4). Write $Z(t) = \int e^{-tK_f}\,d\varpi$ and $E_t[K_f] = Z(t)^{-1}\int K_f e^{-tK_f}\,d\varpi$; here t plays the role of $n\beta$ for a tempered posterior. K_f is bounded on the compact support, so differentiating under the integral gives $(\log Z)' = -E_t[K_f]$. Integrating this and applying (4), for each fixed $t_0 > 0$, with an $O(1)$ term depending on t_0, w^* and μ,

$$\int_{t_0}^{T} E_t[K_f]\,dt = \lambda \log T - (m-1)\log\log T + O(1), \qquad T \to \infty. \tag{12}$$

Dividing by $\log T$ shows that the average of $tE_t[K_f]$ against dt/t tends to λ; the multiplicity enters only at the $\log\log T$ scale. At depth five the endpoint and center placements differ by $1/12$ in λ, while positions 2 and 3 share $\lambda = 7/6$ and differ by $-\log\log T + O(1)$; at depth four all placements share $\lambda = 1$. The averaged quantity therefore separates placements whose exponents differ, but not those that differ only in multiplicity, and only asymptotically. For position 2, with $(7/6, 2)$, against position 1, with $(13/12, 1)$, it orders them correctly only once $\log T > 46$, even ignoring the $O(1)$ term. (The pointwise limit $tE_t[K_f] \to \lambda$ needs finer asymptotics than (4).)

6. Full fibers and global selection

The arrangement theorem describes one realization. The global integral combines the local contributions over the complete target fiber.

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Theorem 3 (Full fibers and global selection). *Assume (3) and $0 \leqslant k \leqslant L$. Let $f = F_{c_k} \circ \cdots \circ F_{c_1}$, with $c_j \neq 0$ and, for even r, $c_j > 0$. For $k = 0$, set $f = x$. In effective coordinates $u = w^r$, the complete fiber consists of the ordered coefficients $c_1, \dots, c_k$ placed in any k positions, with all other coefficients zero; the corresponding raw weights are $c_j^{1/r}$ for odd r and $\pm c_j^{1/r}$ for even r. Its raw fiber $\mathcal{F}_f = \{w : K_f(w) = 0\}$ has size*

$$\#\mathcal{F}_f = \begin{cases} \binom{L}{k}, & r \text{ odd}, \\ 2^k\binom{L}{k}, & r \text{ even}. \end{cases}$$

For every prior satisfying the global conditions of Section 2,

$$\int_{\mathbb{R}^L} e^{-nK_f(w)}\pi(w)\,dw \asymp n^{-[k/2 + A_r(L-k)]}(\log n)^{M_r(L-k)-1}. \tag{13}$$

Repeated coefficients are allowed.

The classification follows by removing outer active blocks successively. For normalized polynomials H, G with constant coefficient zero and linear coefficient one, equality $H + cH^r = G + dG^r$ with $c, d \neq 0$ forces $H = G$ and $c = d$. A leading-coefficient and degree argument proves this recovery step (Appendix D).

The identity loss is coercive, and the $L^2(\mu)$ triangle inequality gives

$$\sqrt{K_f(w)} \geqslant \sqrt{K_0(w)} - \|f - x\|_{L^2(\mu)}.$$

Thus K_f has a positive gap outside small neighborhoods of its finite fiber, making the complementary integral exponentially small. Among local contributions, minimize λ, then maximize m among ties. For $n = L - k$, the inequality $\sum_g (n_g - r)_+ \leqslant (n - r)_+$ gives (13) (Appendix D). The cubic examples have global pairs $(1, 2)$ and $(13/12, 1)$; a prior positive near another realization and supported in a sufficiently small neighborhood of it instead retains its local pair.

7. Formal verification

The pairs, fibers and counts of all three theorems are certified in Lean 4 for localizations on small balls with bounded measurable densities, together with the inverse-power criteria and endpoint divergence; the margin marks link each theorem to its certificate. The transfer to the smooth cutoffs of (4), the zeta-pole interpretation and the two remarks are proved on paper or cited, and meromorphic continuation and leading constants are not certified. Appendix E gives the snapshot, the principal declarations and the audit.

Appendix A. Coefficient reduction, zeros, and coercivity

Write $h_\ell = xT_\ell(x^{r-1})$. The exact recursion

$$T_0(y) = 1, \qquad T_\ell(y) = T_{\ell-1}(y) + u_\ell y T_{\ell-1}(y)^r$$

shows that $h_L - x = \sum_{j=1}^{\nu} P_j(u)x^{d_j}$, where P_j is homogeneous of degree j. Comparing the first two coefficients gives $P_1 = S$ and $P_2 = B = r(S^2 - Q)/2$. At depth one there is no second coefficient, and $B = 0$ because $Q = S^2$.

In a fixed small ball in effective variables, $|P_j(u)| \leqslant C_j \|u\|^j \leqslant C_j Q$ for $j \geqslant 2$. Hence the Gram upper bound gives $K_0 \leqslant C(S^2 + Q^2)$. Conversely, with $\gamma > 0$ the smallest eigenvalue of G,

$$K_0 \geqslant \gamma(S^2 + B^2), \qquad Q = S^2 - 2B/r \leqslant S^2 + 2|B|/r. \tag{14}$$

Locally $S^4 \leqslant CS^2$, so $Q^2 \leqslant C(S^2 + B^2)$. This proves (5). Globally (14) gives

$$Q \leqslant K_0/\gamma + (2/r)\sqrt{K_0/\gamma}.$$

Since $Q = \sum_i w_i^{2r} \to \infty$ as $\|w\| \to \infty$, K_0 is coercive. Also $K_0 = 0$ implies $S = B = Q = 0$, hence $w = 0$. For a finite-mass prior the integral outside any sufficiently small fixed neighborhood is therefore $O(e^{-bN})$ for some $b > 0$.

Appendix B. Angular sectors and the logarithmic factor

For even r, $S \asymp \|w\|^r$ and (5) reduces the integral to an ordinary radial integral with density ρ^{L-1} and loss ρ^{2r}. The pair is $(L/(2r), 1)$. At depth one the loss is exactly a positive constant times w^{2r} for either parity.

Suppose r is odd and $L \geqslant 2$, and use the Euclidean sphere. Let $\tau(\theta) = \sum_i \theta_i^r$, $q(\theta) = \sum_i \theta_i^{2r}$. Compactness gives $0 < c \leqslant q \leqslant C$. The set $\Sigma = \{\tau = 0\}$ is nonempty, since it contains $(2^{-1/2}, -2^{-1/2}, 0, \dots, 0)$. At a point of Σ, Euler's identity gives $\theta \cdot \nabla\tau = r\tau = 0$, whereas $\nabla\tau \neq 0$. Thus the tangential derivative is nonzero. The inverse function theorem supplies smooth local coordinates (t, η) with $t = \tau$, and surface density bounded above and below on smaller coordinate boxes. A finite collection of these boxes covers a neighborhood of the compact set Σ. On the remaining compact angular set, $|\tau|$ has a positive lower bound.

Restrict each chart to the inverse image of a product box $|t| < b$, $\eta \in V$ contained in its image; finitely many smaller boxes cover Σ, and bounded larger boxes give upper bounds. The smaller boxes give lower bounds, while the unused coordinates contribute finite positive volume. For radial cutoff $R > 0$, the product domain lies between $(0, \rho_0) \times (-\rho_0^r, \rho_0^r)$ and $(0, \rho_1) \times (-\rho_1^r, \rho_1^r)$, where $\rho_0 = \min(R, b^{1/r})$ and $\rho_1 = \max(R, b^{1/r})$. These inclusions justify the following model integrals. The comparison (7) is used only on the chart domain itself; the larger boxes carry only the explicit model integrand, so they give valid upper bounds.

The reduced loss equals $\rho^{2r}(t^2 + \rho^{2r}q^2)$ and is comparable to (7). In a product box $0 < \rho < \rho_0$, $|t| < \rho_0^r$, split into two sectors for each sign $\sigma \in \{-1, 1\}$:

$$\begin{aligned}
&\mathcal{C}: \quad t = \sigma\rho^r\nu, \quad 0 < \nu < 1, && \rho^{L-1}\,d\rho\,dt = \rho^{L+r-1}\,d\rho\,d\nu,\\
&\mathcal{O}: \quad t = \sigma z^r, \quad \rho = zy, && \rho^{L-1}|d\rho\,dt| = rz^{L+r-1}y^{L-1}\,dz\,dy,\\
&\qquad 0 < z < \rho_0,\ 0 < y < 1.
\end{aligned}$$

Apart from boundaries of measure zero these are exactly $|t| < \rho^r$ and $\rho^r < |t|$. The loss is comparable to ρ^{4r} in $\mathcal{C}$ and to $z^{4r}y^{2r}$ in $\mathcal{O}$. Sector $\mathcal{C}$ has pair $((L+r)/(4r), 1)$. For $\mathcal{O}$, substitute $a = z^{4r}$ and $b = y^{2r}$. Up to constants its integral is

$$\int_0^{\rho_0^{4r}} \int_0^1 e^{-Nab} a^{\alpha-1} b^{\beta-1} \, db \, da, \qquad \alpha = \frac{L+r}{4r}, \quad \beta = \frac{L}{2r}.$$

The inner integral is comparable to $\min\{1, (Na)^{-\beta}\}$. Splitting at $a = N^{-1}$ gives order $N^{-\min(\alpha,\beta)}$, with a factor $\log N$ exactly when $\alpha = \beta$. The same pair follows directly from the integer monomial $z^{4r}y^{2r}$ with density $z^{L+r-1}y^{L-1}$ by Lin (2017, Proposition 2.1). The noncancellation region contributes $O(N^{-\beta})$, so it does not change the resulting pair. The same integrals with the exponential replaced by the negative power diverge when either exponent reaches its critical value.

Finally, if $cF \leqslant K \leqslant CF$, then the corresponding Laplace integrals satisfy $Z_F(CN) \leqslant Z_K(N) \leqslant Z_F(cN)$ for the same positive amplitude. Thus both the exponent and the logarithmic power transfer to the actual loss. For the pole interpretation, replace the cutoff by a small closed box around the isolated zero, on which the density is smooth and positive. The difference is supported away from the zero and contributes an exponentially small Laplace term and a zeta term holomorphic near the positive real axis. Resolution on that box supplies the stated interpretation.

The Lean certificates localize to balls rather than smooth cutoffs. If $B_{in} \subset \{\chi = 1\}$ and $\operatorname{supp}\chi \subset B_{out}$ are small balls around w^*, then

$$\int_{B_{in}} e^{-NK_f}\pi \, dw \leqslant Z_{N,w^*} \leqslant \|\chi\|_\infty \int_{B_{out}} e^{-NK_f}\pi \, dw.$$

Both outer integrals have the same pair, since any sufficiently small ball does, so Z_{N,w^*} has it too.

For the integrability threshold, Tonelli and $\int_0^\infty t^{s-1}e^{-tK}\,dt = \Gamma(s)K^{-s}$ give, for $s > 0$,

$$\int K_f^{-s}\chi\pi \, dw = \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} Z_{t,w^*} \, dt,$$

because the zero set of K_f meets a small $\operatorname{supp}\chi$ only in the null set $\{w^*\}$. Near $t = 0$ the integrand is integrable for every $s > 0$; at infinity (4) makes it integrable exactly when $s < \lambda$. For $s \leqslant 0$ integrability is trivial.

Appendix C. Proof of the local arrangement theorem

We first identify independent leading coefficient directions. We then control the higher-order terms uniformly, including along cancellation paths. Finally, we transfer the coefficient comparison to the population loss and factor the local integral.

Fix $r, k, (n_g)$ and nonzero active coefficients $c_1, \dots, c_k$. Constants in the coefficient estimates, and the size of the neighborhood, depend only on these fixed data; they may degenerate as some $c_j \to 0$. Use coefficient ℓ^1 norms on polynomial spaces and sum norms on their direct sums; use the Euclidean norm on each effective group vector $u_g = (a_{g,i}^r)_i$. Norm equivalence will be used only in fixed finite dimension.

THE AUXILIARY COMPOSITION MAP. Set $d_h = 1 + h(r-1)$, $\nu_g = (r^{n_g} - 1)/(r-1)$, and

$$\mathcal{P}_g = \operatorname{span}\{x^{d_h} : 1 \leqslant h \leqslant \nu_g\}, \qquad \mathcal{P} = \mathbb{R}^k \oplus \bigoplus_{g=0}^{k} \mathcal{P}_g.$$

Thus $\mathcal{P}_g = \{0\}$ for an empty group and $\mathcal{P}_g = \operatorname{span}\{x^r\}$ for a singleton. For $\nu = (\delta, p_0, \dots, p_k) \in \mathcal{P}$, define

$$\Psi(\nu) = (\mathrm{id} + p_k) \circ F_{c_k+\delta_k} \circ \cdots \circ (\mathrm{id} + p_1) \circ F_{c_1+\delta_1} \circ (\mathrm{id} + p_0). \tag{15}$$

For $k = 0$ this means $\Psi(p_0) = \mathrm{id} + p_0$. This is a polynomial map into the finite-dimensional space of polynomials of degree at most r^L: degrees multiply, and $k + \sum_g n_g = L$. Every output has linear term x, and its other degrees are 1 modulo $r - 1$, so all output differences lie in the sparse feature space of (3). Let $\mathcal{H}_g$ denote the composition of inactive group g. For its error $p_g = \mathcal{H}_g - x$, $\Psi(\nu) = h_L$ and $\Psi(0) = f$.

The differential can be identified exactly. Put $\widehat{H}_0 = x$, $\widehat{H}_j = F_{c_j} \circ \widehat{H}_{j-1}$, and $T_j = F_{c_k} \circ \cdots \circ F_{c_{j+1}}$, with $T_k = \mathrm{id}$. Define the fixed polynomials

$$U_j = (T_j' \circ \widehat{H}_j)\widehat{H}_{j-1}^r, \quad V_g = (T_g' \circ \widehat{H}_g)\widehat{H}_g^r, \quad W_g = (T_g' \circ \widehat{H}_g)\widehat{H}_g^{2r-1}.$$

The chain rule gives, for arbitrary auxiliary group perturbations $q_g \in \mathcal{P}_g$,

$$D\Psi(0)[\alpha, q_0, \dots, q_k] = \sum_{j=1}^{k} \alpha_j U_j + \sum_{g=0}^{k} (T_g' \circ \widehat{H}_g)(q_g \circ \widehat{H}_g). \tag{16}$$

In particular V_g and W_g are the columns for the first two group coefficients when those coefficients exist.

INDEPENDENCE FOR $r \geqslant 3$. First consider the virtual family U_j, V_g, W_g in $\mathbb{R}[x]$, including both directions at every group. At $k = 0$ the family is x^r, x^{2r-1}. Suppose the family at j active blocks is independent and has degrees at most $(2r-1)d$, where $d = r^j$ is the degree of the partial output H. Appending F_e, $e \neq 0$, multiplies all old columns by $P = 1 + reH^{r-1}$ and adds the active column H^r and the last-group columns $\widetilde{H}^r, \widetilde{H}^{2r-1}$, where $\widetilde{H} = H + eH^r$.

The multiplied old degrees are at most $(3r-2)d$; the last two degrees are $r^2 d$ and $r(2r-1)d$, strictly greater and distinct for $r \geqslant 3$. Their coefficients in any relation therefore vanish. The remaining relation is $PR + tH^r = 0$. Since $P - reH^{r-1} = 1$, P and H^r are coprime. The polynomial P is nonconstant, so $P \mid tH^r$ forces $t = 0$, then $R = 0$. This proves independence and preserves the degree bound. Delete both group columns for an empty group and W_g for a singleton. The remaining columns are independent and, by (16), belong to the actual sparse output space. The virtual columns need not belong to that space and are used only to prove independence before restriction.

UNIFORM CONTROL OF OMITTED COEFFICIENTS. Each actual inactive group has

$$\begin{aligned} \mathcal{H}_g &= x + S_g x^r + B_g x^{2r-1} + R_g, \\ B_g &= \frac{r}{2}(S_g^2 - Q_g), \qquad \|R_g\| \leqslant C_g \, \|u_g\|_2 \, Q_g. \end{aligned} \tag{17}$$

Indeed its coefficient of x^{d_h} is homogeneous of degree h in u_g. On $\|u_g\|_2 \leqslant 1$, the sum of the finitely many terms with $h \geqslant 3$ is bounded by $C_g \, \|u_g\|_2^3$, and

$Q_g = \|u_g\|_2^2$. Empty and singleton groups have $R_g = 0$; also $B_g = 0$ for a singleton.

For odd r, let z consist of the active deviations, S_g for $n_g \geqslant 1$, and B_g for $n_g \geqslant 2$. Let J insert these scalars as the corresponding auxiliary coordinates; its norm is one. With

$$\varepsilon = \sum_j |\delta_j| + \sum_g \|u_g\|_2\,, \qquad R = (0, R_0, \ldots, R_k),$$

we have the *exact identity* $v = Jz + R$. Shrink the neighborhood so $\varepsilon \leqslant 1$ and $|S_g| \leqslant 1$ for every group. Since

$$Q_g \leqslant S_g^2 + \frac{2}{r}|B_g| \leqslant |S_g| + \frac{2}{r}|B_g|,$$

(17) gives fixed constants $C_0, C_1 > 0$ such that

$$\|z\| \leqslant C_0\varepsilon, \qquad \|R\| \leqslant C_1\varepsilon\,\|z\|\,. \tag{18}$$

The first bound follows from $|S_g| \leqslant \sqrt{n_g}\,\|u_g\|_2$ and $|B_g| \leqslant C\,\|u_g\|_2^2$. No division by S_g, B_g or $\|z\|$ occurs, so the bounds hold through simultaneous cancellations, including $z = 0$.

Lemma C.1 (Taylor estimate with a controlled remainder)**.** *Let Ψ be a C^2 map between finite-dimensional normed spaces on a neighborhood of the closed unit ball. Let J be linear and suppose $v = Jz + R$, $0 \leqslant \varepsilon \leqslant 1$, and (18) holds. Set $M = \|J\| + C_1$, and assume $MC_0\varepsilon \leqslant 1$. If H_* bounds $\left\|D^2\Psi\right\|$ on that ball, then, with $A = D\Psi(0)J$,*

$$\Psi(v) - \Psi(0) = Az + E, \qquad \|E\| \leqslant \left(\|D\Psi(0)\|\,C_1 + \frac{H_*}{2}M^2C_0\right)\varepsilon\,\|z\|\,. \tag{19}$$

Proof. We have $\|v\| \leqslant M\,\|z\| \leqslant MC_0\varepsilon \leqslant 1$. Taylor's formula with its integral remainder gives the exact decomposition

$$E = D\Psi(0)R + \int_0^1 (1-t)D^2\Psi(tv)[v,v]\,dt.$$

The first term is bounded by $\|D\Psi(0)\|\,C_1\varepsilon\,\|z\|$. The second is bounded by

$$(H_*/2)\,\|v\|^2 \leqslant (H_*/2)M^2\,\|z\|^2 \leqslant (H_*/2)M^2C_0\varepsilon\,\|z\|\,.$$

□

Apply the lemma to (15); a finite H_* exists by compactness because Ψ is polynomial. The retained-column independence gives $a > 0$ with $\|Az\| \geqslant a\,\|z\|$. Let C_* be the parenthesized constant in (19). Choose ε small enough that $MC_0\varepsilon \leqslant 1$ and $C_*\varepsilon \leqslant a/2$, in addition to the preceding group bounds. Then

$$\frac{a}{2}\,\|z\| \leqslant \|h_L - f\| \leqslant (\|A\| + a/2)\,\|z\|\,.$$

All these restrictions hold on a full neighborhood of the fixed raw realization because $\varepsilon \to 0$ there. Norm equivalence, (17), and the sparse Gram bounds give

$$K_f \asymp \sum_j \delta_j^2 + \sum_g (S_g^2 + B_g^2) \asymp \sum_j \delta_j^2 + \sum_g (S_g^2 + Q_g^2).$$

For the last comparison use $Q_g \leqslant S_g^2 + 2|B_g|/r$ and $B_g = r(S_g^2 - Q_g)/2$ with bounded S_g.

Even degree, including $r = 2$. Now $u_g \geqslant 0$, so $Q_g \leqslant S_g^2$. Retain $z = (\delta, S_g : n_g \geqslant 1)$ and set $R'_g = \mathcal{H}_g - x - S_g x^r$. Homogeneity of all omitted coefficients, now of degree at least two, gives

$$\left\|R'_g\right\| \leqslant C_g Q_g \leqslant C_g S_g^2 \leqslant C_g \sqrt{n_g}\, \varepsilon S_g.$$

Thus $\nu = Jz + (0, R'_0, \ldots, R'_k)$ obeys the same lemma. The retained family U_j, V_g is independent by the previous induction with old degree bound rd, multiplied bound $(2r-1)d$, and new last-group degree $r^2 d > (2r-1)d$ for every $r \geqslant 2$; coprimality separates the new active column. Absorption yields $K_f \asymp \sum_j \delta_j^2 + \sum_g S_g^2$. Finally

$$\sum_{i \text{ in } g} |w_i|^{2r} \leqslant S_g^2 \leqslant n_g \sum_{i \text{ in } g} |w_i|^{2r}$$

for each nonempty group, proving (10).

Integration. The Jacobian determinant of the active coordinate map is $\prod_j r(w_{i_j}^*)^{r-1} \neq 0$. Its inverse absolute Jacobian and the prior are bounded above and away from zero locally. Inner and outer product boxes in $(\delta, a_0, \ldots, a_k)$ bound the cutoff integral. On each box the comparison integral factors into Gaussian active integrals and identity-group integrals. Theorem 1, or the usual addition rule (Lin, 2017, Proposition 3.7), gives (11): exponents add and $m - 1 = \sum_g (M_r(n_g) - 1)$. All comparison losses vanish only at the base point, so shrinking cutoffs does not change the pair.

Appendix D. Full fibers and global selection

Ordered recovery. Every active block has degree r, so a network with q active blocks has degree r^q. Equal outputs therefore have the same active count. To recover their coefficients, suppose

$$H + cH^r = G + dG^r, \qquad c, d \neq 0,$$

where $H(0) = G(0) = 0$ and $H'(0) = G'(0) = 1$. Equality of output degrees gives $\deg H = \deg G = D \geqslant 1$. Choose $a \neq 0$ so that $U = aG$ and H have the same leading coefficient. Comparing leading output coefficients gives $d = ca^r$, hence

$$c(H - U) \sum_{i=0}^{r-1} H^i U^{r-1-i} = G - H.$$

The sum has degree $(r-1)D$: its leading coefficient is r times the $(r-1)$st power of the common leading coefficient. If $H - U \neq 0$, its zero constant coefficient forces degree at least one. The left side would then have degree greater than D, while the right side has degree at most D. Thus $H = U$; the linear coefficients give $a = 1$, and consequently $H = G$ and $c = d$. Iterating this step recovers the ordered active chain.

Fiber size. The positive-definite feature Gram makes zero population loss equivalent to polynomial equality. The recovered chain can occupy any k of the L positions, in its fixed order. Each placement has one raw realization for odd r and 2^k independent sign choices for even r. Different supports or signs give different vectors, even when active coefficients repeat. This proves the stated count.

Global selection. Coercivity from Appendix A and the L^2 triangle inequality make every sublevel set of K_f compact. Delete disjoint small neighborhoods of the finite fiber. Choose a large ball outside which $K_f \geqslant 1$. Within the ball, the complement of the open fiber neighborhoods is compact and zero-free, so continuity gives a positive minimum whenever that complement is nonempty. Set $b = 1$ if that complement is empty; otherwise take the smaller of its minimum and one. Then $K_f \geqslant b > 0$ on the entire remaining set, so the prior-weighted tail is at most $e^{-bN} \int \pi$. Continuity and positivity of the prior at each fiber point give local upper and positive lower bounds. Each fiber neighborhood therefore has the pair of Theorem 2 with a positive contribution. A finite sum of terms $N^{-\lambda_i}(\log N)^{m_i-1}$ has the smallest λ_i and then the largest m_i among ties. Several realizations with that same pair affect the multiplicative constants, not the logarithmic power; multiplicities do not add across disjoint neighborhoods.

For even r, every placement has pair $(k/2+(L-k)/(2r), 1)$. For odd r, write $n = L-k = \sum_g n_g$. Then

$$\sum_g A_r(n_g) = \frac{n}{2r} - \frac{1}{4r}\sum_g (n_g - r)_+, \qquad \sum_g (n_g - r)_+ \leqslant (n-r)_+.$$

If $n < r$, all placements have the same exponent and $m = 1$. If $n = r$, all exponents agree and concentrating the inactive blocks in one gap gives $m = 2$, the largest multiplicity. If $n > r$, the inequality is strict whenever at least two gaps are nonempty: if any gap exceeds r, the other nonempty gaps strictly reduce the sum of positive parts from $n - r$; if none does, that sum is zero. Hence concentration in one gap uniquely describes the minimizing patterns, and these have $m = 1$. Such a placement exists before, between, or after the ordered active blocks. This proves (13).

Appendix E. Lean source and audit record

The proof source is available at

https://github.com/bubskee/residual-slt-in-a-haystack

The reproducibility snapshot, containing the proof source and audit, is

`b08b1a334abac97cbdf754ba14eacee82e10be70`

It uses Lean `v4.35.0` and mathlib commit

`56a005855ea17e0b0958f73ec801ae140b0a1679`

The principal declarations, all in namespace `ResidualSLT`, are

- `unified_residual_network_certificate`: full fibers, exact counts, and local/global learning pairs;
- `sparse_identity_network_rlct_laplace_certificate`: the identity pair;
- `arrangement_raw_network_rlct_laplace_certificate_of_depth_pos`: the local arrangement pair;
- `network_full_raw_fiber`: ordered active-chain classification;
- `raw_network_fiber_cardinality`: exact raw-fiber size.

Exact hypotheses are recorded in `paper/formalization_status.md`. From the repository root, run:

```
git checkout b08b1a334abac97cbdf754ba14eacee82e10be70
cd lean
lake exe cache get
lake build
lake env lean -DwarningAsError=true -DautoImplicit=false \
  ../paper/checks/NetworkAudit.lean
```

The audit prints the certificate definitions and the types and axiom dependencies of these five declarations. Every dependency set is exactly `propext`, `Classical.choice`, and `Quot.sound`; none contains `sorryAx`. The logs are in `paper/checks/`: `lean-build.txt` and `NetworkAudit.txt`.

Tool and computational resource disclosure. Following the disclosure recommendation of the Leiden Declaration on Artificial Intelligence and Mathematics (Alper et al., 2026), we report substantial AI assistance. Astra and Opus 5.5 assisted with writing, proof development, and Lean formalization. Fable 5.1 provided adversarial review and attempted to find errors in the arguments. The author retains responsibility for the results, arguments, and citations.

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Independent researcher